

Physics Origins 05: Toward the Schrödinger Equation

Part 1 — Vector Geometry from First Principles

September 23, 2026

This is the prerequisite chapter of the Schrödinger series. The main thread is the shear transformation: how one geometric operation becomes one matrix. Along the way it settles the inner product, the cross product, the angle-difference formula, the rotation matrix and the Pythagorean theorem, and closes with the axiomatisation of area and Russell’s paradox. One question runs through all of it — **how does a sentence about geometry turn into an expression in algebra?**

Contents

1	Where this chapter sits	3
2	Shear: from a geometric definition to a matrix	3
2.1	Three geometric requirements	3
2.2	Why the displacement is kb and not k	4
2.3	The geometric meaning of k : $k = \tan \varphi$	5
2.4	The change-of-basis sandwich: $A = QSQ^T$	6
2.5	Area preservation: causal direction and algebraic characterisation	6
2.6	Physical origin and algebraic status	7
3	The inner product: projection, componentwise products, and a common slip	7
3.1	“The inner product is the projection” is imprecise	7
3.2	Why cos rather than sin	8
4	The cross product and the area of a parallelogram	8
4.1	Two derivations of the area formula	9
4.2	Deriving $S = a b \sin \theta$ itself	9
4.3	Signed area and the determinant	9
5	The angle-difference formula: one route through a geometric jigsaw	10
5.1	The construction	10

5.2	The lengths of the pieces	10
5.3	Finishing by substitution	11
5.4	Why the complex-number route is circular	11
5.5	The historical order	11
6	The rotation matrix: projection onto a new basis	12
6.1	The row view	12
6.2	Active versus passive	13
6.3	The same structure as shear	13
7	The Pythagorean theorem and <i>Elements</i> I.47	13
7.1	The configuration and the goal	13
7.2	A route using only I.41	13
7.3	Comparison with Euclid's own proof	15
7.4	Euclid's linguistic constraints	15
7.5	Other standard proofs	16
7.6	The <i>Elements</i> itself	16
8	What area is: a formula that took four millennia to prove	16
8.1	Axiom or theorem	16
8.2	The earliest appearances: an algorithm, not a theorem	17
8.3	Why the Greeks did not prove it	17
8.4	Completion at the end of the nineteenth century	17
9	Axiom systems and Russell's paradox	18
9.1	The rule in naive set theory	18
9.2	The paradox, formalised	18
9.3	Two different granularities: $\in$ and $\subseteq$	18
9.4	What was refuted was the rule, not that definition	19
9.5	ZFC: how the hole is plugged	19
9.6	Two independence results	19
9.7	One family of structures	20
9.8	Three unrelated " $1 + 1$ "s	20
9.9	An assessment	20
10	Summary	21

1 Where this chapter sits

This piece did not start from shear. It is a by-product of working backwards from the Schrödinger equation.

The chain runs like this. To write the Schrödinger equation you need de Broglie's postulate $\lambda = h/p$; to make sense of that postulate you need to know where Planck's constant came from; Planck's constant comes out of black-body radiation and the quantisation of the electromagnetic field, which sends you back to Maxwell's equations; and the differential form of Maxwell's equations rests on curl and divergence.

At the curl-and-divergence layer what remains is no longer physics. It is vector geometry itself: inner products, cross products, orthonormal bases, linear maps. These are usually handed over as definitions in class, but a definition is a model of something, not a gift from the sky. Part 1 works at that layer.

Shear is the thread because it is simple enough to hold in one hand and still uses every ingredient of the layer: a fixed line, a normal direction, a projection, a change of basis, a determinant. Once shear is fully understood, the inner product and rotation come almost for free.

2 Shear: from a geometric definition to a matrix

2.1 Three geometric requirements

The definition of shear has only three clauses. The algebraic formula is what they force, not the other way round.

Given a line L through the origin, with unit direction vector $\mathbf{u}$ and unit normal $\mathbf{n}$:

1. every point of L stays put;
2. every other point moves **parallel to** L , i.e. along $\mathbf{u}$;
3. the displacement is **proportional to that point's signed distance to** L , with constant of proportionality k .

The phrase "signed distance to L " in clause 3 can only be expressed algebraically by the normal inner product: $\mathbf{n} \cdot \mathbf{x}$ is 0 on L , grows in magnitude with distance, and changes sign across L . Putting the three together,

$$\mathbf{x}' = \mathbf{x} + k(\mathbf{n} \cdot \mathbf{x})\mathbf{u} \quad (1)$$

The division of labour is worth memorising: $\mathbf{n}$ sets *how much*, $\mathbf{u}$ sets *which way*. The two must be different directions. If the displacement were taken along $\mathbf{n}$, points would leave L perpendicularly and points *on* L would be pushed off it, breaking clause 1 immediately.

The simplest case takes L to be the x -axis ($\mathbf{u} = (1, 0)$, $\mathbf{n} = (0, 1)$):

$$x' = x + ky, \quad y' = y, \quad A = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$$

Along $y = x$ we have $\mathbf{u} = \frac{1}{\sqrt{2}}(1, 1)$ and $\mathbf{n} = \frac{1}{\sqrt{2}}(-1, 1)$; substituting into (1),

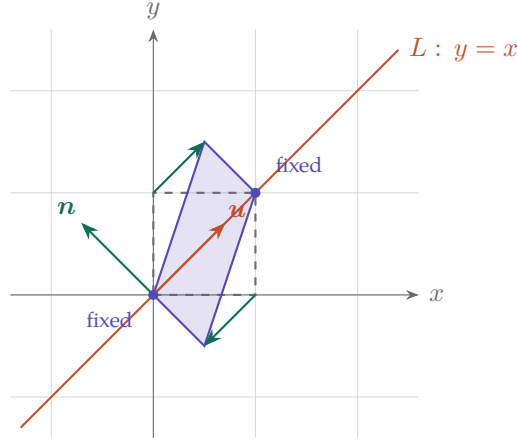


Figure 1 Shear along $L : y = x$ with $k = 1$. The dashed outline is the original square. The two vertices on L , $(0,0)$ and $(1,1)$, do not move; the other two lie on opposite sides and slide along $\pm \mathbf{u}$ by an amount proportional to their distance from L . Area is unchanged.

$$A = \begin{bmatrix} 1 - \frac{k}{2} & \frac{k}{2} \\ -\frac{k}{2} & 1 + \frac{k}{2} \end{bmatrix}, \quad \det A = \left(1 - \frac{k}{2}\right)\left(1 + \frac{k}{2}\right) + \frac{k^2}{4} = 1$$

The two are the same thing in different axes, which §2.4 makes explicit.

2.2 Why the displacement is kb and not k

This is the line separating shear from translation, and the easiest place to slip.

Change to the $\{\mathbf{u}, \mathbf{n}\}$ frame and write

$$a = \mathbf{u} \cdot \mathbf{x} \text{ (position along } L), \quad b = \mathbf{n} \cdot \mathbf{x} \text{ (distance from } L)$$

In this frame shear reads

$$a' = a + kb, \quad b' = b \tag{2}$$

It is tempting to write $a' = a + k$ instead. That does keep b fixed, but it moves every point by the same amount, which is a rigid translation along $\mathbf{u}$. Testing a point of L exposes it: shearing along $y = x$, take $(1,1)$, for which $a = \sqrt{2}$ and $b = 0$. Under $a' = a + k$ it slides along $y = x$, whereas shear demands that it stay exactly where it is.

Translating “a stack pushed over but not squashed” into four requirements leaves (2) as the only option:

1. **not squashed:** layers neither approach nor separate, so heights are preserved and $b' = b$;
2. **whole layers translate:** points at the same b move identically, so the displacement is a function of b alone, $a' = a + f(b)$;
3. **bottom layer pinned:** the layer L does not move, so $f(0) = 0$;
4. **linear:** the map must be expressible as a matrix, so $f(b) = kb$.

Clause 4 is there purely so that a matrix exists. If f were allowed to be arbitrary, say $f(b) = b^2$, the first three clauses would still hold — layers still translate rigidly, the bottom one still pinned — but the map would be nonlinear and area need not be preserved.

Looking at a single point cannot distinguish kb from k , since for fixed b they differ by a constant factor. The difference only shows up across points at different b , which is exactly the contrast in the next figure.

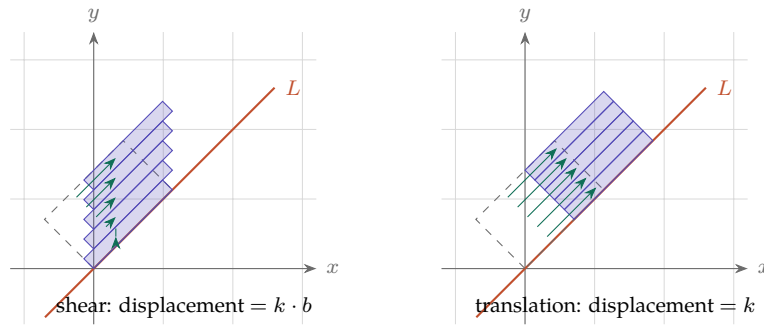


Figure 2 Left: displacement grows linearly with height, the layer on L stays put, and the shape is pushed over. Right: every layer moves by the same amount, including the one on L ; the shape is unchanged and simply relocated. Dashed outlines mark the original position.

2.3 The geometric meaning of k : $k = \tan \varphi$

k is not a coefficient pulled out of nowhere. It is the tilt of the edge that got pushed over.

Take an edge originally perpendicular to L . After the shear it becomes a slanted line making angle φ with its original position. Any point on it has moved b along $\mathbf{n}$ and kb along $\mathbf{u}$, so

$$\tan \varphi = \frac{kb}{b} = k$$

Two equivalent readings:

- **tilt**: $k = \tan \varphi$. For small deformations $k \approx \varphi$; in engineering k is the **shear strain**, written γ .
- **slope**: every line originally perpendicular to L becomes one of a family of parallel slanted lines, whose slope relative to $\mathbf{n}$ is k .

So what kb means on the page is this: **the ratio of displacement to distance-from- L is constantly k** , independent of the point. The b only says how far along that slanted line the point has travelled.

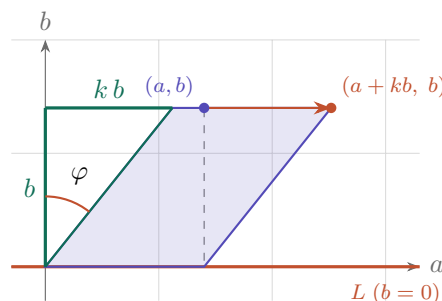


Figure 3 Seen in the $\{\mathbf{u}, \mathbf{n}\}$ frame. The small triangle at the left is where k comes from: vertical leg b , horizontal leg kb , angle φ , and $\tan \varphi = k$. Any point moves from a to $a + kb$ with its b untouched.

2.4 The change-of-basis sandwich: $A = QSQ^T$

A shear in any direction is “change basis $\rightarrow$ do the easy thing $\rightarrow$ change back”. Three steps:

1. **project**: resolve x onto $\{u, n\}$, giving $a = u \cdot x$ and $b = n \cdot x$;
2. **shear**: apply the simplest form in the new coordinates, $a' = a + kb$, $b' = b$;
3. **write back**: $x' = a'u + b'n$.

Expanding step 3 returns the definition:

$$a'u + b'n = (a + kb)u + bn = (au + bn) + kb u = x + k(n \cdot x)u$$

In matrix form,

$$A = QSQ^T, \quad Q = [u \ n], \quad S = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix} \quad (3)$$

Q^T projects, S shears in the new frame, Q writes the result back.

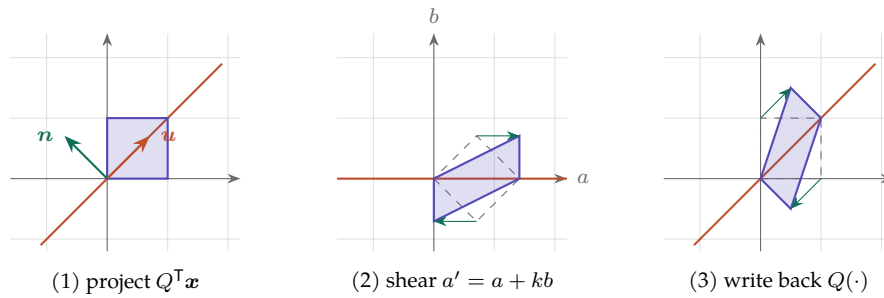


Figure 4 The three steps of $A = QSQ^T$. In the middle panel L has been straightened into the horizontal axis and the shear collapses to its simplest form.

There is a visual trap here: the middle panel looks like a rotation. That is because Q is a rotation matrix (45° when L is $y = x$). It comes from the change of coordinates and is cancelled by Q^T at the end. **The two slices of bread are always rotations; the filling decides what the whole thing is:**

filling S	whole map $A = QSQ^T$
$\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$	shear
$\text{diag}(\lambda_1, \lambda_2)$	independent scaling along u and n
a rotation	still a rotation (it commutes with Q , which cancels)

The same pattern reappears everywhere: diagonalisation, SVD and PCA are all this sandwich.

2.5 Area preservation: causal direction and algebraic characterisation

Area preservation is a consequence of the definition, not its motivation. The converse fails: rotations preserve area, and so does stretching by λ in one direction while compressing by $1/\lambda$ in another.

The chain runs: points of L fixed + every line parallel to L translated rigidly $\Rightarrow$ every slice keeps its length $\Rightarrow$ area unchanged. One can see it by slicing (Cavalieri's principle) or simply compute $\det A = 1$.

Algebraically, which linear maps are shears? Write $N = A - I$, the displacement map. The condition is

$$N \neq 0, \quad N^2 = 0$$

Intuitively, N sends any point to a displacement along L , and points of L are sent to 0 by N ; applying it twice must therefore vanish. For the x -axis shear, N sends (x, y) to $(ky, 0)$, and a second application gives 0 because the y -component is already zero.

In two dimensions this is equivalent to $\text{tr } A = 2$ together with $\det A = 1$ (and $A \neq I$). $\det = 1$ **alone is not enough** — rotations satisfy it too. The extra condition $\text{tr} = 2$ encodes the structure “one whole line fixed pointwise, everything else sliding parallel to it”.

2.6 Physical origin and algebraic status

The definition comes from continuum mechanics; mathematicians did not invent it a priori. The prototype is a deck of cards: hold the bottom card down, push the top, and each card slides within its own plane by an amount proportional to its height above the bottom.

quantity in the definition	its counterpart in the deck
L	the bottom card, held fixed
$\mathbf{u}$	the direction the cards slide
$b = \mathbf{n} \cdot \mathbf{x}$	height above the bottom
k	shear strain γ : slip per unit height

This is exactly “simple shear” in the mechanics of materials, with displacement field $u_x = \gamma y$; the name goes back to scissors, whose two blades slide past one another. The fluid counterpart is Couette flow: two parallel plates, the lower fixed and the upper dragged at constant speed, with the fluid flowing in layers whose speed grows linearly with height. After time t one has $k = \dot{\gamma}t$ — which is precisely the picture in the layered figure above.

Area preservation carries physical content here: layers only slide relative to one another, without compression or extension, so volume (area in two dimensions) is conserved. That is what distinguishes shear deformation from compressive deformation.

Algebraically the map has another name, **transvection**. Every matrix of determinant 1 factors into a product of shears, and the elementary row operation “add a multiple of one row to another” in Gaussian elimination *is* a shear. So it is a basic building block in algebra as well, not merely a physical model.

3 The inner product: projection, componentwise products, and a common slip

3.1 “The inner product is the projection” is imprecise

That statement is only strictly true when $|\mathbf{a}| = 1$.

The definition of projection has already divided out $|a|$:

$$\text{length of the projection of } \mathbf{b} \text{ onto } \mathbf{a} = |\mathbf{b}| \cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}|}$$

It depends on the direction of $\mathbf{a}$ only, never its length. The inner product, by contrast, is

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta = |\mathbf{a}| \times (\text{projection length})$$

The two get conflated because the commonest situation is projection onto unit vectors — the axes $\hat{x}, \hat{y}$, or any orthonormal basis — where $a_x = \mathbf{a} \cdot \hat{x}$ and the two numbers genuinely coincide.

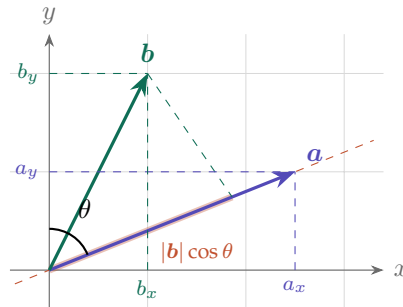


Figure 5 The thick orange segment is the projection of $\mathbf{b}$ onto the direction of $\mathbf{a}$, of length $|\mathbf{b}| \cos \theta$. The inner product equals $|\mathbf{a}|$ times this length; the two agree only when $|\mathbf{a}| = 1$.

3.2 Why $\cos$ rather than $\sin$

The angle θ ranges over $[0, \pi]$. On that interval:

- $\cos \theta$ decreases monotonically from 1 to -1 : 1 for aligned, 0 for perpendicular, -1 for opposed — exactly a scale of “how aligned”;
- $\sin \theta$ is not monotone, and $\sin \theta = \sin(\pi - \theta)$: aligned and opposed both give 0, and 30° and 150° give the same value, so it cannot tell alignment from opposition.

The deeper reason is the decomposition of $\mathbf{b}$ into a component along $\mathbf{a}$ and a component perpendicular to it.

	component taken	what it measures	vanishes when
inner product $ \mathbf{a} \mathbf{b} \cos \theta$	along $\mathbf{a}$	alignment, signed	perpendicular
cross product $ \mathbf{a} \mathbf{b} \sin \theta$	perpendicular to $\mathbf{a}$	spread (area)	parallel

That $\cos$ is monotone while $\sin$ is not is no coincidence: it reflects the difference between “along” and “across”. The component along $\mathbf{a}$ carries a sign and so distinguishes same from opposite; the perpendicular component only records how far off the line of $\mathbf{a}$ the vector sits, regardless of which side.

4 The cross product and the area of a parallelogram

4.1 Two derivations of the area formula

The formula $S = a_x b_y - a_y b_x$ need not be assumed; either route derives it.

Route one: Lagrange's identity. From $S = |\mathbf{a}||\mathbf{b}| \sin \theta$ and $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos \theta$,

$$\begin{aligned} S^2 &= |\mathbf{a}|^2 |\mathbf{b}|^2 - (\mathbf{a} \cdot \mathbf{b})^2 \\ &= (a_x^2 + a_y^2)(b_x^2 + b_y^2) - (a_x b_x + a_y b_y)^2 \\ &= (a_x b_y - a_y b_x)^2 \end{aligned}$$

On expansion the terms $a_x^2 b_x^2$ and $a_y^2 b_y^2$ cancel, leaving a perfect square.

Route two: use shear itself. This presupposes that shear preserves area, which must be argued by slicing — not by the area formula, or the reasoning would be circular. Assume $a_x \neq 0$:

1. shear along y , $(x, y) \mapsto (x, y - \frac{a_y}{a_x} x)$, flattening $\mathbf{a}$ into $(a_x, 0)$ and sending $\mathbf{b}$ to $(b_x, b_y - \frac{a_y b_x}{a_x})$;
2. shear along x to push $\mathbf{b}$ upright; this step does not change its height.

The result is a rectangle of base a_x and height $b_y - \frac{a_y b_x}{a_x}$, hence area $a_x b_y - a_y b_x$.

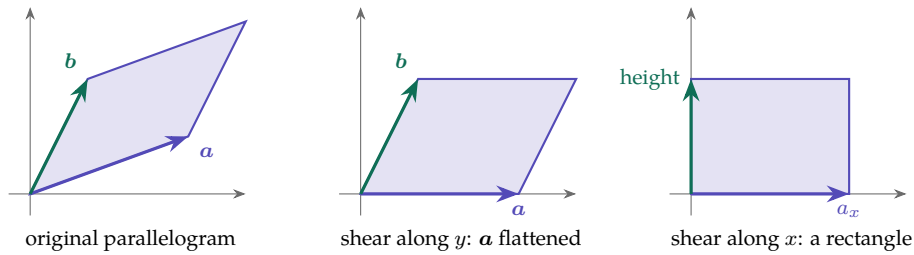


Figure 6 Two shears turn the parallelogram into a rectangle of equal area. Each step preserves area, so $S = a_x b_y - a_y b_x$.

4.2 Deriving $S = |\mathbf{a}||\mathbf{b}| \sin \theta$ itself

This formula is not given either. Two steps:

Step one: area of a parallelogram = base $\times$ height. Take $\mathbf{a}$ as the base and drop a perpendicular from the tip of $\mathbf{b}$ to the line of $\mathbf{a}$. That cuts off a right triangle; translate it to the other side and the figure closes into a rectangle with the same base and height. This cut-and-paste move is itself a shear — pushing the slanted edge upright without changing area.

Step two: height = $|\mathbf{b}| \sin \theta$. The height is the perpendicular distance from the tip of $\mathbf{b}$ to the line of $\mathbf{a}$. The vector $\mathbf{b}$, that perpendicular, and the projection of $\mathbf{b}$ along $\mathbf{a}$ form a right triangle with hypotenuse $|\mathbf{b}|$, and the leg opposite θ is the height. The definition of $\sin$ finishes it. When θ is obtuse the relevant angle inside the triangle is $\pi - \theta$, and since $\sin(\pi - \theta) = \sin \theta$ the formula is unchanged.

4.3 Signed area and the determinant

Embed $\mathbf{a}$ and $\mathbf{b}$ in three dimensions as $(a_x, a_y, 0)$ and $(b_x, b_y, 0)$; the z -component of their cross product is $a_x b_y - a_y b_x$.

This quantity is **signed**: positive when $\mathbf{b}$ lies counterclockwise from $\mathbf{a}$, negative when clockwise. Strictly it is a **signed area**, and the geometric area is its absolute value. The signed quantity is precisely the 2×2 determinant — the same $\det$ as in $\det A = 1$ earlier. “Area is preserved” really says that signed area is preserved.

5 The angle-difference formula: one route through a geometric jigsaw

5.1 The construction

On the unit circle take two angles α and β , assuming for now $0 < \beta < \alpha < \pi/2$:

- B is the point on the unit circle at angle β , and D is its foot on the x -axis;
- A is the foot of the perpendicular from B to the ray at angle α , and E is the foot of A on the x -axis;
- C is where the horizontal line through A meets BD .

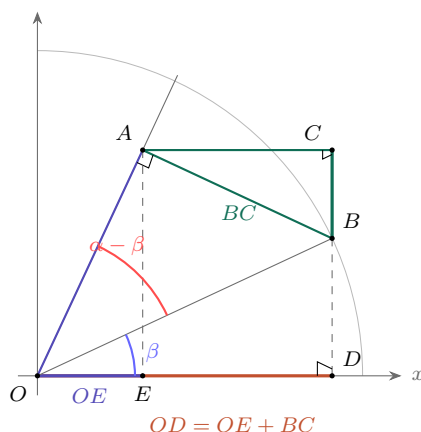


Figure 7 The jigsaw behind the angle-difference formula. Here $OB = 1$ and A is the foot of the perpendicular from B to the ray at angle α . The base OD splits into OE and ED , and $ED = BC$.

5.2 The lengths of the pieces

Since $OB = 1$,

$$OD = \cos \beta, \quad BD = \sin \beta$$

A is the projection of B onto the ray at angle α , of length $|OB| \cos(\alpha - \beta)$, so

$$OA = \cos(\alpha - \beta)$$

Projecting OA onto the x -axis in turn,

$$OE = OA \cos \alpha = \cos(\alpha - \beta) \cos \alpha$$

The segment AB is the perpendicular from B to that ray, of length $AB = \sin(\alpha - \beta)$. In the right triangle ABC — with AC vertical, BC horizontal and the right angle at C — we have $AB \perp OA$ while OA makes angle α with the x -axis, so $\angle BAC = \alpha$ and

$$BC = AB \sin \alpha = \sin(\alpha - \beta) \sin \alpha$$

Both BC and ED are horizontal segments between $x = OE$ and $x = OD$, hence $ED = BC$. From $OD = OE + ED$,

$$\cos \beta = \cos(\alpha - \beta) \cos \alpha + \sin(\alpha - \beta) \sin \alpha \quad (4)$$

5.3 Finishing by substitution

Equation (4) already has the right shape; only the variables sit in the wrong places. Put $\beta' = \alpha - \beta$, i.e. $\beta = \alpha - \beta'$:

$$\cos(\alpha - \beta') = \cos \beta' \cos \alpha + \sin \beta' \sin \alpha$$

Multiplication commutes and β' is only a name, so renaming it back to β :

$$\boxed{\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta} \quad (5)$$

One gap remains open: this picture only covers the configuration $0 < \beta < \alpha < \pi/2$. Extending to arbitrary angles needs a separate argument, or a case reduction using the reflection identities.

5.4 Why the complex-number route is circular

The natural thought is: since $r(\cos \theta + i \sin \theta)$ multiplies by “moduli multiply, arguments add”, doesn’t the addition formula follow at once?

That route is circular. Complex multiplication is defined by an algebraic rule,

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i$$

and showing that its geometric meaning is “moduli multiply, arguments add” is exactly where the addition formula gets used. So deriving the addition formula from polar multiplication goes in a circle and returns to its starting point — unless the argument-addition property is established independently, for instance via Euler’s formula, where $e^{i\theta}$ is defined by its power series and the addition theorem follows from multiplying series, with no trigonometric geometry involved.

5.5 The historical order

Geometry came first, complex numbers later, vectors last:

date	event	note
c. AD 150	Ptolemy's table of chords	equivalent of the sum/difference formulas, via the cyclic-quadrilateral theorem
16th c.	Cardano, Bombelli	complex numbers forced out by cubic equations; purely algebraic, no geometry
1797–1806	Wessel, Argand	the complex plane: a geometric reading of complex numbers
1840s–1880s	Hamilton → Gibbs, Heaviside	quaternions to vector analysis; inner and cross products take shape

So the order is: trigonometric geometry → complex numbers as algebra → complex numbers as geometry → vectors and inner products. Complex numbers were not invented to be consistent with geometry; they were forced out by equation solving, and only two centuries later was their agreement with existing trigonometry verified.

As an aside on teaching: Chinese high-school texts run the reverse route — first define $a \cdot b = |a||b| \cos \theta$ geometrically and derive the coordinate formula (distributivity being justified by “projections add”, which uses no trigonometric identity, so no circularity), then use the coordinate formula to recover $\cos(\alpha - \beta)$. The two routes are inverse to each other and both are sound.

6 The rotation matrix: projection onto a new basis

6.1 The row view

$$R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

The two row vectors

$$\mathbf{r}_1 = (\cos \theta, -\sin \theta), \quad \mathbf{r}_2 = (\sin \theta, \cos \theta)$$

form an orthonormal basis, obtained by rotating the standard basis **clockwise** by θ . The components of Rx are just $\mathbf{r}_1 \cdot x$ and $\mathbf{r}_2 \cdot x$: **project x onto this basis, write the two resulting coordinates as components in the original frame, and the point you get is x rotated counterclockwise by θ .**

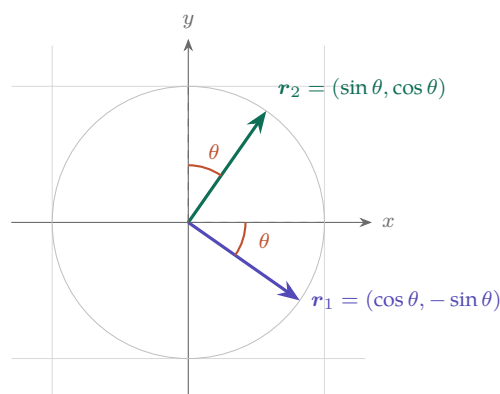


Figure 8 The two rows of the rotation matrix. They are the standard basis vectors rotated clockwise by θ , while points are rotated counterclockwise by θ .

6.2 Active versus passive

Basis clockwise, point counterclockwise: that sign flip is the classic confusion between the **active** view (the point moves, the frame stays) and the **passive** view (the frame moves, the point stays). One matrix, three ways of talking:

view	statement	usual habitat
columns (active)	the columns are the images of e_1, e_2 ; Ax combines them	linear algebra texts
rows (passive)	each row is dotted with x , giving one projected coordinate	signal processing: DFT, wavelets,
frame change (passive)	the frame rotated, the point did not	physics and robotics: attitude ma

Courses usually emphasise the column view and rarely spell out the active/passive distinction, which is why these three threads are seldom tied together.

6.3 The same structure as shear

The shear sandwich $A = QSQ^T$ and rotation share one structure: change basis, do something simple, change back. Only the filling differs. Rotation is special in that its filling commutes with Q , so the sandwich collapses to a single layer — which is why the change-of-basis step is invisible there.

The row view also gives a quick check. The rows of R are orthonormal, so $R^T R = I$ and $\det R = \pm 1$; continuity ($\det = 1$ at $\theta = 0$) forces $\det R = 1$. Area is preserved, yet $\text{tr } R = 2 \cos \theta \neq 2$ unless $\theta = 0$, so rotation is not a shear — which is the content of “ $\det = 1$ is not enough” from §2.5.

7 The Pythagorean theorem and *Elements* I.47

7.1 The configuration and the goal

Triangle ABC has its right angle at A . On the three sides, squares $BDEC$, $ABFG$ and $ACKH$ are erected outwards. Drop $AL \perp BC$ and extend it to meet DE at M , splitting the large square into rectangles $BDML$ and $LMEC$.

The goal is $ABFG = BDML$; the other half, $ACKH = LMEC$, is symmetric and copies over verbatim.

In modern notation the goal reads $AB^2 = BL \cdot BD$. But that is a later algebraic transcription, not Euclid’s language — see §7.4.

7.2 A route using only I.41

The proposition used throughout:

I.41 If a parallelogram has the same base as a triangle and they lie between the same parallels, the parallelogram is double the triangle.

This is far more permissive than “cut along the diagonal”: the triangle’s apex need not be a vertex of the parallelogram, only somewhere on that parallel line. Slide it anywhere along the line — same base, same height.

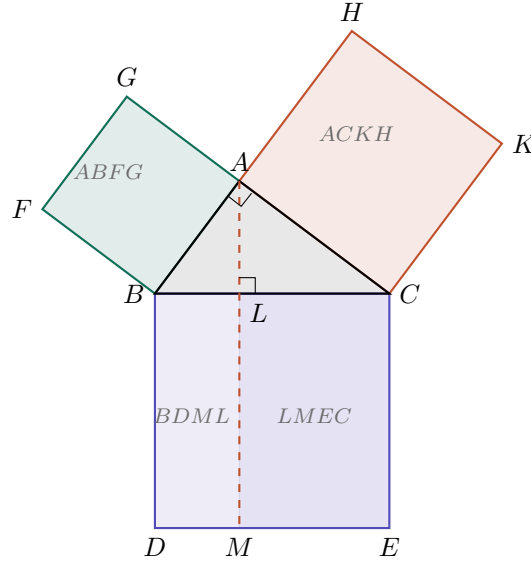


Figure 9 The I.47 configuration. $AL \perp BC$ extended to M on DE cuts $BDEC$ into $BDML$ and $LMEC$. The target is $ABFG = BDML$.

The chain below uses I.41 only, with no congruence:

1. Extend MA to meet the line FG at X . Since $AB \parallel FG$, triangle ABX has base AB with apex on the parallel through FG , so by I.41

$$\triangle ABX = \frac{1}{2} ABFG$$

2. Triangles BDA and BDX share the base BD , and both A and X lie on the line through A parallel to BD (namely XM), so

$$\triangle BDA = \triangle BDX$$

3. Triangles BDX and BAX share AX as a common base, with the distance from B to AX as their common height, so

$$\triangle BDX = \triangle BAX$$

4. Triangle BDA has base BD with apex A on the line LM , so by I.41

$$\triangle BDA = \frac{1}{2} BDML$$

Chaining them:

$$\frac{1}{2} BDML = \triangle BDA = \triangle BDX = \triangle BAX = \frac{1}{2} ABFG \implies ABFG = BDML$$

Two gaps left open. First, the collinearity of X, A, M is read off the picture; a rigorous treatment needs I.14 (if two lines meet a third at a point on opposite sides with adjacent angles summing to two right angles, they form one straight line), the same tool used to show GA and AH form a straight line. Second, the right half $LMEC = ACKH$ simply copies the left.

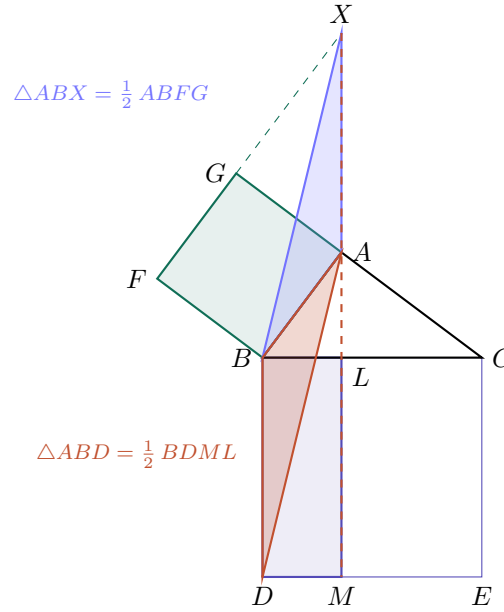


Figure 10 The equal-area chain. X falls on the extension of FG (green dashed), outside the segment itself, which by I.41 does not affect the conclusion. The blue and orange triangles are halves of the two figures, bridged through $\triangle BD X$.

7.3 Comparison with Euclid’s own proof

Euclid introduces no X and goes through congruence instead:

By I.14, GA and AC form one straight line. Then $\triangle ABD \cong \triangle FBC$ by I.4 (two sides and the included angle: $AB = FB$, $BD = BC$, equal included angles). Triangle ABD shares base BD with $BDM L$ between the same parallels, and $\triangle FBC$ shares base FB with the square $ABFG$; applying I.41 twice to halve each gives $ABFG = BDM L$.

	the route above	Euclid’s proof
propositions used	I.41 (twice)	I.4 + I.41
new points	X , far outside the figure	none
extra burden	prove X, A, M collinear	prove GA, AC form a straight line
reading experience	all area, one continuous chain	one congruence step inserted

Counted strictly the two are comparable; neither is clearly simpler. The route above has the advantage of using a single tool throughout (area), at the cost of a point X so far away that the figure becomes awkward to draw.

7.4 Euclid’s linguistic constraints

One point is easy to forget today: **Euclid had no numerical notion of area**. The *Elements* contains no formula of the type “area = base $\times$ height”, only relations of equality and multiple between figures.

- AB^2 can only be phrased as “the square on AB ”;
- $BL \cdot BD$ can only be phrased as “the rectangle contained by BL and BD ”;

- nothing may be transposed, cancelled, or divided through.

The entire argument has to be carried by statements like “this figure equals that one” and “this is double that”. His toolkit at this point is just I.4 (SAS congruence), I.14 (forming a straight line), I.41 (the doubling relation), plus the common notion that equals added to equals give equals.

7.5 Other standard proofs

Several hundred proofs are known; three more are worth recording:

- **Dissection** (the easiest to reconstruct yourself): inside a square of side $a + b$, place four congruent right triangles. One arrangement puts one in each corner, leaving a square of side c ; the other pairs them into rectangles against two sides, leaving squares of sides a and b . The empty area is the same in both, so $c^2 = a^2 + b^2$. Only additivity of area is used.
- **Similar triangles**: drop the altitude from the right angle to the hypotenuse; both small triangles are similar to the original, giving $a^2 = cp$ and $b^2 = cq$, which add to the result. This is the proof closest to the inner-product/projection thread.
- **Not allowed**: $\sin^2 + \cos^2 = 1$. That identity is the Pythagorean theorem rewritten, so using it is circular.

7.6 The *Elements* itself

Written around 300 BC by Euclid in Alexandria, under Ptolemy I. The Greek original is long lost; current editions descend mainly from Theon’s fourth-century recension, together with an earlier manuscript found in the Vatican in the nineteenth century. The first six books were translated into Chinese in 1607 by Matteo Ricci and Xu Guangqi — the Chinese terms for “geometry”, “point”, “line” and “parallel” were fixed then — and the remaining nine had to wait until 1857, when Li Shanlan and Alexander Wylie completed them.

8 What area is: a formula that took four millennia to prove

8.1 Axiom or theorem

“The area of a square is the side squared” is often treated as an axiom. In fact it is a **theorem**. What is genuinely assumed is something else:

- the unit square has area 1 (a convention, not provable);
- area is additive (non-overlapping pieces add);
- area is invariant under translation and rotation.

From these three: for integer side n , cut the square into n^2 unit cells and count; for rational sides, subdivide and reduce to the integer case; for irrational sides, approximate from inside and outside and pass to the limit.

So, precisely: “the unit square has area 1” is a convention; “a square of side a has area a^2 ” is a theorem derived from it.

8.2 The earliest appearances: an algorithm, not a theorem

Long before Euclid, and in every case as a computational rule:

date	source	content
c. 1800 BC	Babylonian tablets	square fields by multiplying the side by itself; YBC 7289 gives $\sqrt{2}$ in sexagesimal
c. 1650 BC	Rhind papyrus (Egypt)	square fields likewise; circular fields approximated via $8/9$ of the diameter
c. 1st c. AD	<i>Jiuzhang Suanshu</i> , ch. 1	“square times itself” as the area rule; root extraction as its inverse

All of them arise from dividing land into unit cells and counting. It is empirical generalisation, with no notion of proof — and none needed.

8.3 Why the Greeks did not prove it

The Greeks did not prove this, because **within their framework it cannot even be stated**: area is not a number, and a segment cannot be “multiplied by itself” into an area.

What they did instead was prove proportions. *Elements* VI.19 and VI.20 state that similar figures are to one another in the duplicate ratio of corresponding sides, i.e. $S_1/S_2 = (a_1/a_2)^2$. Note that what appears is the **square of a ratio**: ratios are dimensionless and may be compounded, whereas segments may not. That distinction is strict in their hands.

The proposition rests on Eudoxus’s theory of proportion (*Elements* Book V), which handles incommensurable magnitudes and is therefore untroubled by irrationals. From “similar squares are as the squares of their sides”, choosing one square as a unit gives the modern formula; but that final step requires the notion that area *is* a number, which the Greeks decline to take.

8.4 Completion at the end of the nineteenth century

Turning it into a theorem derived from axioms had to wait for the theory of real numbers and measure:

- **Peano (1887), Jordan (1892)**: content defined by inner and outer approximation with small cells, from which the area a^2 of a square of side a follows;
- **Lebesgue (1902)**: the complete version. The measure of a rectangle is defined as length times width, and one then proves that the outer measure constructed from this, restricted to rectangles, still equals length times width — a step with real content, not a circularity — together with countable additivity;
- **Hilbert (1899)**, *Grundlagen der Geometrie*: a different road, building a theory of area inside a purely geometric axiom system without recourse to real numbers.

The timeline, then: four thousand years as a computational rule → two thousand years as a statement about proportions of figures → definition and proof only at the end of the nineteenth century. Supplying rigorous foundations for propositions “too obvious for anyone to think of proving” is characteristic of that period’s foundational movement.

9 Axiom systems and Russell's paradox

9.1 The rule in naive set theory

Late nineteenth-century set theory had one entirely natural rule — written down explicitly by Frege as Basic Law V in the *Grundgesetze*, and used naively by Cantor:

Unrestricted comprehension For any property P there is a set $\{x \mid P(x)\}$ consisting exactly of the objects satisfying P .

This rule underpins the whole logicist programme of reducing mathematics to pure logic. Frege's aim was a foundation **without exceptions**.

9.2 The paradox, formalised

In 1901 Russell took $P(x)$ to be “ x is not a member of itself”, forming

$$R = \{x \mid x \notin x\}$$

For a general x , $x \in R$ yields no contradiction whatever: by definition $x \in R \iff x \notin x$, and both sides agree. The contradiction appears only on substituting $x = R$:

$$R \in R \iff R \notin R$$

If the left side holds so does the right; if the right side holds the left fails. Neither branch survives.

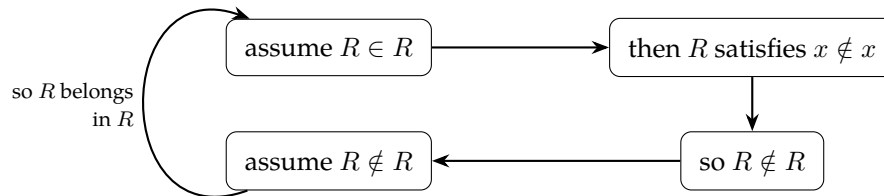


Figure 11 The self-referential loop. Both directions return to the negation of their own assumption. Note that substituting $x = R$ is a legitimate preliminary step rather than a conclusion, and that one may not presuppose whether R satisfies the condition — that is precisely what is at issue.

9.3 Two different granularities: $\in$ and $\subseteq$

English “contains” blurs two relations, and here they must be kept apart:

- $x \subseteq x$ (a set is a **subset** of itself): true of every set, trivial, harmless;
- $x \in x$ (a set is a **member** of itself): the whole set appears as one element inside itself — a completely different granularity.

For example $\{1, 2\} \subseteq \{1, 2\}$ holds, while $\{1, 2\} \in \{1, 2\}$ does not, since its members are only 1 and 2. Russell's paradox uses the latter.

9.4 What was refuted was the rule, not that definition

This is the point most easily stated backwards. **Russell did not prove that the definition of R is illegitimate; he proved that the set-forming rule is untenable.** Under the rules of the day R was perfectly legitimate; the fault lay in the rule that produced it.

The distinction decides the magnitude of the consequences. Had it merely been one pathological definition, striking it out would have settled the matter. What actually happened is that the law generating it collapsed and the system had to be rebuilt. Frege received the letter while volume III was in press and conceded in an appendix that the foundation had given way.

The same period produced Cantor's paradox (the set of all sets, whose power set is larger) and the Burali-Forti paradox (the set of all ordinals), all pointing at the same defect. Russell's is the most economical, hence the most famous.

9.5 ZFC: how the hole is plugged

Zermelo (1908) proposed axioms restricting how sets may be formed; Fraenkel and Skolem (1922) completed the system, giving ZF, and with the axiom of choice, ZFC. Its strategy is to **say nothing about what a set is, and only regulate which sets may be built.**

axiom	role
extensionality	sets with the same members are equal
separation	subsets may only be carved out of existing sets, never conjured from a property
pairing, union, power set	build new sets from given ones
infinity	an infinite set exists
replacement	the image of a set under a function is a set
foundation	excludes pathologies such as $x \in x$
choice	a choice function exists for any family of non-empty sets

What blocks Russell's paradox is **separation**: R can no longer be formed, because there is no set of all sets to carve it out of.

Worth stressing separately: the case $x \in x$ is excluded by the **axiom of foundation**, independently, and has nothing to do with Russell's argument. In other words, the intuition that " $x \in x$ looks suspicious" was never vindicated by that argument — its conclusion concerns the rule. The suspicious self-membership was dealt with separately, later.

9.6 Two independence results

Once ZFC was in place, two natural questions turned out to have no answer inside it:

- **the axiom of choice** is independent of ZF: Gödel (1938) showed it cannot be refuted, Cohen (1963) that it cannot be proved, using forcing;
- **the continuum hypothesis** is independent of ZFC: the same two authors supplied the two halves.

Moreover, by Gödel's second incompleteness theorem, ZFC cannot prove its own consistency.

9.7 One family of structures

Russell's self-referential construction is neither isolated nor marginal. The same family produced several load-bearing theorems of the twentieth century:

theorem	year	form of self-reference
Cantor's diagonal argument	1891	build an object differing from every row of the list
Gödel's incompleteness	1931	the arithmetic encoding of "this sentence is unprovable"
Tarski's undefinability	1933	truth cannot be defined within the system
Turing's halting problem	1936	feed the deciding program to itself

Gödel and Turing turned this self-reference into a **tool** rather than treating it as a disease.

9.8 Three unrelated "1 + 1"s

A cluster of confusions worth clearing up in passing:

phrase	what it actually is
$1 + 1 = 2$ in Peano arithmetic	2 is defined as the successor of 1; follows from the recursion for +
$1 + 1 = 2$ in <i>Principia</i>	cardinal version: 1 is the class of singletons, addition is disjoint union
Chen Jingrun's " $1 + 2$ "	Goldbach: every large even number is a prime plus a number with at most two prime factors

The second entry appears only in volume II because type theory, relations, equinumerosity and cardinals all have to be erected first. The proposition itself is not hard; what is hard is the programme of constructing number out of pure logic. That programme was later judged a failure — type theory and the axiom of reducibility were unnatural, and Gödel's 1931 result closed it off — but its technical legacy is enormous, and modern mathematical logic grew largely out of it.

The third entry has nothing to do with arithmetic as written: the 1 and 2 count prime factors of the two summands. Goldbach's conjecture itself is the " $1 + 1$ " case and remains open.

9.9 An assessment

A natural objection: working mathematicians never used sets that way, so was Russell not simply nitpicking?

The first half is right. Analysis, algebra and geometry never touch such constructions, and no existing theorem was invalidated when the paradox appeared. But "nitpicking" is the wrong description, for two reasons:

1. **The goal was different.** A system claiming universal validity is finished by a single counterexample; "the other 99.99% works" is no defence. A contradiction in a formal system proves every proposition (*ex falso quodlibet*), so the system becomes worthless. That is not the situation of an engineering tool that misbehaves in a corner case.
2. **The corner turned out to be the main road.** All four theorems in the table above grow from the same family of structures. The point that was picked out happened to be the most fertile direction available.

Besides, Russell wrote that letter precisely because he endorsed Frege's programme and wanted to advance it. Testing a system that claims to be exception-free by hunting for exceptions is the job, not an intrusion.

The real disagreement is whether the programme was worth undertaking at all. Poincaré mocked set theory as a disease that later generations would study as such; Kronecker had opposed Cantor earlier still. Historically each side was half right: the programme failed, but computability theory, model theory and the independence methods it generated became disciplines of their own.

10 Summary

Collecting the planar linear maps that appeared in this chapter:

map	matrix (standard position)	det	tr	what stays fixed
identity	I	1	2	the whole plane
shear	$\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$	1	2	a line L , pointwise
rotation	$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$	1	$2 \cos \theta$	the origin only
equiareal scaling	$\begin{bmatrix} \lambda & 0 \\ 0 & 1/\lambda \end{bmatrix}$	1	$\lambda + 1/\lambda$	the origin only

All four preserve area, but only shear satisfies $N^2 = 0$ for $N = A - I$. **det = 1 speaks about area; tr = 2 speaks about structure.** This is the most compact instance of the question this chapter opened with:

the sentence in geometry	where it lands in algebra
signed distance to the line L	$n \cdot x$
push along L	multiply by u
points of L stay fixed	displacement vanishes at $b = 0$, hence $= kb$
the push grows uniformly	linearity, hence $f(b) = kb$ and not b^2
do the same thing in another direction	$A = QSQ^T$
area is preserved	$\det A = 1$
one line fixed pointwise	$\text{tr } A = 2$ and $N^2 = 0$

Part 2 returns to physics: from the orthonormal bases and projections built here into the differential structure of fields — curl and divergence — and on up to Maxwell's equations.

The two digressions in this chapter — the axiomatisation of area, and Russell's paradox with ZFC — are not on the physical path. But they answer the same kind of question: when, and by what means, does something long used as "obvious" finally get proved? That question runs through the whole Origins series.